The second-order asymptotic of the EKG sequence

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Abstract

The EKG sequence starts $a(1) = 1$, $a(2) = 2$, and thereafter takes the least unused positive integer not coprime to its predecessor. In 2002, Lagarias, Rains and Sloane conjectured $a(n) = n + n/(3\log n) + o(n/\log n)$ for almost all n. Replacing values $p, 3p$ by $2p$ for odd primes p defines $b(n)$; we prove, at every index n, $b(n) = n + n/(3\log n) + o(n/\log n)$ as $n \to \infty$ (log natural). Writing $E(t)$ for the least positive even integer outside $\{a(1), \ldots, a(t)\}$, Haver reduced its asymptotic to counting used odd values above $E(t)$ other than $3p$ (p odd prime), and proved the sharp upper bound $E(t) - t \leq (1/3 + o(1))t/\log t$; we prove the lower half and passage to all indices. For $x = E(t)$ as $t \to \infty$, every $a(i)$ $(1 \leq i \leq t)$ other than $3p$ (p odd prime) is at most $x + o(x/\log x)$, and every unused composite $h < x$ satisfies $x - h = o(x/\log x)$ uniformly in h. The proofs inject selected neighbours of even terms into prime cofactors whose positive density in a common interval contradicts an elementary sieve bound. Haver's counting identity and the prime number theorem yield $1/3$. The complete Lean formalization imports Haver's estimates and the prime number theorem as proved theorems.

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1 Introduction

The EKG sequence, first studied by Lagarias, Rains, and Sloane [7], starts with $1, 2$ and then chooses the least unused positive integer having a common factor with the preceding term. It is A064413 in the On-Line Encyclopedia of Integer Sequences [9]. Write $U_t = \{a(1), \ldots, a(t)\}$. The rule is

$$a(1) = 1, \quad a(2) = 2, \quad a(n) = \min\{m \geq 1 : m \notin U_{n-1},\ \gcd(m, a(n-1)) > 1\} \quad (n \geq 3). \tag{1}$$

Its first terms are $1, 2, 4, 6, 3, 9, 12, 8, 10, 5, 15, 18, 14, 7, 21$. We use LRS numbering from the arXiv text; in the journal version their Conjectures 1–3 are Conjectures 4.1–4.3. Lagarias, Rains, and Sloane prove the permutation property, increasing order of the primes (their Lemma 2), and linear upper and lower bounds. Their numerical experiments suggested three branches: ordinary values near n, primes near $n/2$, and three times primes near $3n/2$. Their PDF abstract says, exactly, "We conjecture that almost all $a(n)$ satisfy the asymptotic formula" and displays

$$a(n) = n(1 + 1/(3\log n)) + o(n/\log n) \quad \text{as } n \to \infty.$$

Their Conjecture 2 defines $\sim$ as a ratio tending to 1 and gives these exact mathematical expressions, with the indicated conditions:

$$\begin{aligned} a(n) &\sim n[1 + 1/(3\log n)], && a(n) \neq p, 3p \ (p \text{ prime}), \\ a(n) &\sim \tfrac{1}{2}n[1 + 1/(3\log n)], && a(n) = p \ (p \text{ prime}), \\ a(n) &\sim \tfrac{3}{2}n[1 + 1/(3\log n)], && a(n) = 3p \ (p \text{ prime}). \end{aligned}$$

Thus the displayed Conjecture 2, read literally, asks only for three ratio limits. We interpret its intended second-order content in the stronger additive sense stated in their PDF abstract. The arXiv web abstract has different parentheses; the PDF is the source here.

Hofman and Pilipczuk [6, Theorem 4.3] prove that every odd prime p is preceded by $2p$; the following $3p$ is the consequence already noted in LRS Conjecture 1. Their proof uses the LRS computation for $p \leq 25000$; Haver's proof reduces this finite range [5]. Hofman and Pilipczuk also prove the three ratio limits in LRS Conjecture 2: their Theorem 4.17 gives $a'_n = n + O(n/\log\log\log n)$. Their a'_n also replaces the prime 2, so it agrees with our $b(n)$ only up to finitely many indices and yields the same asymptotic bound. Introducing Conjecture 5.1, they call the LRS formula the "stronger conjecture", supporting the additive second-order reading above. Haver [5] proves the stronger $b(n) = n + O(n/\log n)$ and quantitative estimates for early even values, used values above the least unused even integer, and missing composites below it. Haver's Corollary 2(e), in version 3 dated October 4, 2026, reduces the second-order assertion at these even integers to a negligible count of odd used values above them. Haver's Theorem 1(d) and Corollary 12(d) already give $E(t) - t \leq (1/3 + o(1))t/\log t$ and $N_p \geq 2p - (2/3 + o(1))p/\log p$, with sharp constants. Our contribution is the lower half of the clock, $E(t) - t \geq (1/3 - o(1))t/\log t$, and the all-index transfer. We prove the remaining hypothesis in Haver's Corollary 2(e), and a uniform bound for missing composites needed for that transfer. The reduction and the estimates it uses are Haver's.

For comparison with (1), the Lean construction in `EKG/Canonical.lean` is transliterated below (`Nat` denotes the natural numbers). It is zero-based internally; `chooseNext` selects the least positive unused eligible value, with existence and minimality proved in `EKG/Basic.lean`.

```
noncomputable def actualA (n : Nat) : Nat :=
  if n = 0 then 1 else if n = 1 then 2 else
  chooseNext (fun i : Fin n => actualA i.val)
```

```
    (actualA (n-1))
termination_by n
decreasing_by all_goals omega
```

The one-based interface in `EKG/EndpointDefinitions.lean` uses `a n := EKG.canonical.a (n - 1)`. Its value at zero is an unused total extension. Section 8 explains the checked existence, gcd eligibility and minimality statements, not additional assumptions on this construction.

We use the smoothing convention of LRS Figure 5 [7]:

$$b(n) = s(a(n)), \qquad s(m) = \begin{cases} 2p, & m = p \text{ or } m = 3p \text{ for an odd prime } p, \\ m, & \text{otherwise.} \end{cases} \tag{2}$$

In particular, $s(2) = 2$ and $s(6) = 6$. OEIS A256417 includes the prime 2 in both replacements [9]; it differs from b only at indices $2, 4$, where our values are $2, 6$ and its values are $4, 4$. Here is the corresponding Lean definition, transliterated from `EKG/EndpointDefinitions.lean`, using ASCII type names:

```
def smoothValue (m : Nat) : Nat :=
  if And m.Prime (Not (m = 2)) then 2 * m
  else if And (m / 3).Prime
      (And (Not (m / 3 = 2)) (m = 3 * (m / 3))) then
    2 * (m / 3)
  else m
noncomputable def b (n : Nat) : Nat := smoothValue (a n)
```

Here `And` and `Not` are Lean's logical constructors; the distributed source uses the usual symbolic notation. All logarithms are natural.

Theorem 1. *For the sequence defined by* (1) *and* (2),

$$b(n) = n + \frac{n}{3\log n} + o\left(\frac{n}{\log n}\right) \qquad (n \longrightarrow \infty).$$

Precisely, for every real $\varepsilon > 0$ *there is an* $N \in \mathbb{N}$, $N \geq 3$, *such that for every* $n \in \mathbb{N}$ *with* $n \geq N$,

$$\left| b(n) - n - \frac{n}{3\log n} \right| \leq \varepsilon \frac{n}{\log n}. \tag{3}$$

The Lean type, transliterated using `abs`, `And` and `Exists` for the usual logical notation and opening `EKG.CanonicalEndpoint`, is

```
theorem ekg_sharp_all_index :
  forall eps : Real, 0 < eps ->
    Exists (fun N : Nat => And (3 <= N)
```

```
(forall n : Nat, N <= n ->
  abs ((b n : Real) - (n : Real) -
    (n : Real) / (3 * Real.log (n : Real))) <=
  eps * ((n : Real) / Real.log (n : Real))))
```

The theorem has no sequence-estimate hypotheses. The separate statement in `Statement.lean` and its type check in `Check.lean` fix exactly these quantifiers and casts.

Corollary 2 (The three branches)**.** *Along the indices of each indicated kind, as $n \to \infty$,*

$$\begin{aligned} a(n) &= n + \frac{n}{3\log n} + o(n/\log n), && a(n) \neq p, 3p \quad (p \text{ odd prime}),\\ p &= \frac{n}{2} + \frac{n}{6\log n} + o(n/\log n), && a(n) = p,\\ 3p &= \frac{3n}{2} + \frac{n}{2\log n} + o(n/\log n), && a(n) = 3p. \end{aligned}$$

Proof. In the three cases $a(n) = b(n)$, $a(n) = b(n)/2$, and $a(n) = 3b(n)/2$, respectively. Multiply Theorem 1 by the fixed factor. The finitely many cases involving 2 do not matter for any limit. □

This proves the additive second-order interpretation, and in particular implies the literal Conjecture 2, whose ratio limits were already proved by Hofman and Pilipczuk. Among the three numbered LRS conjectures, Conjecture 3 remains open: $13n/28 \le a(n) \le 12n/7$ globally, with equality only at $n = 28$ and $n = 7$, respectively. It is not settled here; an eventual asymptotic does not settle the required finite range. Nor do we address the unproved distributional suggestions for higher-order fluctuations in LRS Section 4 or their unnumbered cycle questions.

The coefficient has the counting heuristic of LRS Section 4 [7]:

$$n \approx x - \frac{x}{2\log x} + \frac{x}{6\log x} = x - \frac{x}{3\log x}.$$

Our new step upgrades two $O(x/\log x)$ displacement bounds to little-o. For the upper bound, the old greedy inequality is $D \le 2rk(t)$, with $r \le \sqrt{e+D}$ and $k(t) \le K_0\sqrt{e}/\log e$. Multiplying these estimates leaves a fixed coefficient of $e/\log e$; it does not force that coefficient to vanish. For the lower bound, $D \le 2\,\mathrm{spf}(h)F(e)$ and $\mathrm{spf}(h) \le \sqrt{e}$ have the same loss. Haver already identifies these rough-predecessor and rough-missing-composite obstructions in the remark after Corollary 12 [5]. The additional information is sparsity among the forced prime cofactors. Given a putative displacement $D \ge \delta x/\log x$, we instead select multiples $2rMj$ and force their distinct neighbours to be products rq_j with q_j prime. There are at least $D/(4rM) - O(1)$ such cofactors in an interval of length $O(x/(r\log x))$. For each fixed $\delta > 0$ their density is bounded below by a positive constant independent of x. It need not be independent of δ: M also depends on δ. The sieve bound permits an arbitrarily small fixed upper density, chosen after δ and M, which is exactly what the little-o conclusion needs.

The two estimates are an upper bound for the largest ordinary used value above the least unused even integer, and a lower bound on every unused composite below it. Sections 4 and 5

prove them, then exact counting and the prime number theorem give the result. We extract no explicit remainder rate. The nonquantitative step is the choice of a fixed sieve modulus for each δ, followed by unspecified eventual cutoffs in localization and the qualitative prime number theorem. This is not a claim of an intrinsically ineffective obstruction: the sieve constants are computable, but this proof does not supply a function $N(\varepsilon)$. An explicit rate looks within reach using Brun–Titchmarsh in place of the fixed-modulus sieve; this is left for future work.

2 Frontiers and established estimates

Write $L(x) = x/\log x$ for $x > 1$, and let $U_t = \{a(1), \dots, a(t)\}$. Let $E(t)$ be the least positive even integer outside U_t. Call a value $3p$, for an odd prime p, a *tick*, and put

$$T(t) = \max\{a(i) : 1 \le i \le t,\ a(i) \text{ is not a tick}\}, \qquad K(t) = \#\{m \in U_t : 2 \mid m,\ m > E(t)\}.$$

The frontier $E(t)$ is nondecreasing and tends to infinity, by the permutation property. Here and below, $\pi(x)$ counts primes at most the real number x, and $\mathrm{spf}(h)$ is the least prime factor of h. We use the following previously established estimates.

Proposition 3 (Established sequence estimates)**.** *There are fixed constants $C \ge 1$ and $C_1 > 0$ such that, eventually, writing $e = E(t)$ and $k(t) = K(t) + 1$,*

$$|b(n) - n| \le CL(n), \tag{4}$$

$$k(t) \le K_0 \frac{\sqrt{e}}{\log e}, \qquad K_0 = 793, \tag{5}$$

$$T(t) - e \le C_1 L(e). \tag{6}$$

If $t \ge 9$ and a composite $15 \le h < e$ is outside U_t, then

$$e - h \le 2\,\mathrm{spf}(h)\Big(13\pi(\sqrt{h}) + 724\Big). \tag{7}$$

For all sufficiently large e,

$$F(e) := 13\pi(\sqrt{e}) + 724 \le B\frac{\sqrt{e}}{\log e}, \qquad B = 802. \tag{8}$$

Every composite below 15 is in U_{13}. For each odd prime p, its first multiple occurs at an index N_p, and

$$a(N_p) = 2p = E(N_p - 1), \quad a(N_p + 1) = p, \quad a(N_p + 2) = 3p. \tag{9}$$

Distinct introduction indices N_p are at least four apart.

These are results about the actual greedy sequence, not hypotheses added to Theorem 1. The prime pattern goes back to Hofman and Pilipczuk [6]; we use its introduction formulation as proved in Haver's archive [5]. The early-even bound follows from Haver's bound $K(t) \le 12\pi(\sqrt{e}) + 720$ and elementary prime estimates. One may take $C_1 = 4 \cdot 793^2 + 2 \cdot 793$ in (6). The printed sources of (7) and (4) are Haver's Theorem 11 and Corollary 12(c), respectively. The constants in (5)–(8) are convenient weakenings proved in the accompanying sources, rather than optimized constants: $793 = 12 \cdot 6 + 721$ and $802 = 13 \cdot 6 + 724$ follow from the eventual bound $\pi(\sqrt{e}) \le 6\sqrt{e}/\log e$ and $\sqrt{e}/\log e \ge 1$, absorbing the additive constants. The prime estimate in (8) does not require the prime number theorem.

3 Greedy bounds and interval prime capacity

If q is any prime divisor of $a(j-1)$, every positive multiple of q smaller than $a(j)$ was used before index j; otherwise the greedy choice would be smaller. Consequently, at any observation time t,

$$a(j) \le E(t) + 2qk(t) \qquad (3 \le j \le t+1). \tag{10}$$

Indeed, all multiples of $2q$ strictly between $E(t)$ and $a(j)$ are used even values above $E(t)$, and there are at least $(a(j) - E(t))/(2q) - 1$ of them. Their number is at most $K(t)$. Notice that (10) holds for every prime divisor q, at the same observation time, even if j is much earlier than t.

The following elementary fixed-modulus sieve estimate is included with its proof; compare the sieve of Eratosthenes in [1].

Lemma 4. *For every $\eta > 0$ there is a constant C_η such that every closed real interval of length $z \ge 0$ contains at most $\eta z + C_\eta$ primes.*

Proof. Choose a positive integer $Z \ge 2$ sufficiently large, and put $W = \prod_{p \le Z} p$. Unique factorization and geometric series give

$$\frac{W}{\varphi(W)} = \prod_{p \le Z} (1 - 1/p)^{-1} \ge \sum_{m=1}^{Z} \frac{1}{m} \ge \log(Z+1).$$

We may therefore choose Z with $\varphi(W)/W < \eta$. A residue class modulo W has at most $z/W + 1$ members in such an interval. Primes not dividing W lie in its $\varphi(W)$ reduced residue classes, and at most $\omega(W)$ primes divide W. Thus $C_\eta = \varphi(W) + \omega(W)$ works. Both constants are fixed before the interval varies. □

Lemma 5 (Forced common factor)**.** *Let $x, \beta > 0$, let r be prime, and let $M \ge 1$ be an integer. Suppose every prime divisor of $2M$ is smaller than $\beta\sqrt{x}$, and $(\beta\sqrt{x})^3 > 2x$. For selected positive integers $j < \beta\sqrt{x}$, put $y_j = 2rMj$. Suppose the distinct integers d_j are composite, satisfy $d_j < 2x$, have every prime divisor at least $\beta\sqrt{x}$, and satisfy $\gcd(d_j, y_j) > 1$. Then $d_j = rq_j$ for distinct primes q_j. If all d_j belong to an interval I, the q_j belong to the interval I/r.*

Proof. Choose a prime ℓ dividing both d_j and $2rMj$. Every prime divisor of j is at most $j < \beta\sqrt{x}$; every prime divisor of $2M$ is smaller than that bound by assumption. Since all prime divisors of d_j are at least $\beta\sqrt{x}$, ℓ divides neither j nor $2M$. Euclid's lemma forces $\ell \mid r$, so $\ell = r$. A composite d_j has at least two prime factors counted with multiplicity. It cannot have three, since their product would be at least $(\beta\sqrt{x})^3 > 2x > d_j$. It therefore has exactly two, one of which is r; the other is a prime q_j, with $q_j = r$ allowed. Distinct d_j give distinct q_j by cancellation of the fixed positive integer r. Division by r gives the interval assertion. □

We also record how the old rate localizes adjacent ordinary values. If $y = a(j)$ is unchanged by smoothing and $e/2 < y < 2e$, then, uniformly as $e \to \infty$, the occurrence index j tends to infinity. To see this, choose any fixed index cutoff and exceed the maximum of the finitely many preceding values. Equation (4) then gives $j/2 \le y \le 3j/2$ and hence $e/3 < j < 4e$. If an adjacent value $d = a(j-1)$ or $a(j+1)$ is also unchanged by smoothing, the old rate at both indices gives

$$|d - y| \le CL(j) + CL(j \pm 1) + 1 = O(L(e)). \tag{11}$$

The constant is uniform. This uses only the old big-O rate, not the sharp clock we will prove.

4 The upper excursion

Proposition 6. *As $t \to \infty$, $\max\{T(t) - E(t), 0\} = o(L(E(t)))$.*

Proof. Fix $\delta > 0$. All constants in this proof are chosen before the final threshold on t. Write $e = E(t)$, $v = T(t)$, and suppose

$$D = v - e \ge \delta L(e). \tag{12}$$

The old span implies $v < 2e$ for sufficiently large e. Choose $i \le t$ with $a(i) = v$, and let $c = a(i-1)$. The predecessor is not even: the unused even e would then be an eligible choice smaller than v. It is not 1, which appears only at the first index, and it is not an odd prime, since v would then be a tick. If it were a tick, (10) with $q = 3$ would give $D \le 6k(t) = o(L(e))$, a contradiction. Thus c is an odd composite non-tick, so $c \le v$. For $r = \mathrm{spf}(c)$, (10) and (5) imply

$$\alpha\sqrt{e} \le r \le \sqrt{v} < \sqrt{2e}, \qquad \alpha = \frac{\delta}{2B}. \tag{13}$$

Here $B \ge K_0$, so the displayed lower bound is a weakening of $r \ge \delta\sqrt{e}/(2K_0)$.

Set $\beta = \delta/(4K_0)$, and fix an integer $M \ge 2$ with $M > 1/(\alpha\beta)$. Consider

$$Y = \{2rMj : j \in \mathbb{N},\ j \ge 1,\ e + D/2 < 2rMj < v\}.$$

Every member of Y is a multiple of r smaller than v, so it occurred before i. Counting a lattice in an open interval gives

$$|Y| \geq \frac{D}{4rM} - 2. \tag{14}$$

For $y = a(i_y) \in Y$, put $d_y = a(i_y - 1)$. For every prime $q \mid d_y$, the common observation-time inequality (10) yields

$$q > \frac{D}{4k(t)} \geq \beta\sqrt{e}.$$

For large e this excludes 2 and 3. The predecessor is not 1, a prime, or a tick: in the prime case its successor would be odd, whereas y is even. Therefore d_y is an odd composite non-tick, and $d_y \leq v < 2e$. For large e, $(\beta\sqrt{e})^3 > 2e$.

Write $y = r(2Mj)$. Equation (13) gives

$$j < \frac{\sqrt{e}}{\alpha M} < \beta\sqrt{e}.$$

For large e, every prime factor of the fixed integer $2M$ is also smaller than $\beta\sqrt{e}$. Consecutive values d_y, y share a prime, and distinct selected values have distinct predecessor indices and hence distinct predecessors. Lemma 5 therefore gives

$$d_y = rq_y \quad \text{for distinct primes } q_y. \tag{15}$$

Both y and d_y are unchanged by smoothing. The finite-prefix argument preceding (11) applies uniformly to their indices. Since $e < y < v \leq e + C_1 L(e)$, it gives, for a fixed $A > 0$,

$$e - AL(e) \leq d_y \leq e + AL(e).$$

For example, $A = C_1 + 8C + 1$ suffices. Distinct y have distinct occurrence indices, distinct predecessor indices, and hence distinct d_y. Thus (15) injects Y into the primes in $[(e - AL(e))/r, (e + AL(e))/r]$. By Lemma 4,

$$\frac{D}{4rM} - 2 \leq \eta \frac{2AL(e)}{r} + C_\eta.$$

Choose $\eta = \delta/(32AM)$ after fixing M. It follows that

$$D \leq \frac{\delta}{4} L(e) + 4\sqrt{2}M(C_\eta + 2)\sqrt{e}.$$

Since $\sqrt{e} = o(L(e))$, the right side is less than $\delta L(e)$ for sufficiently large e, contradicting (12). The frontier tends to infinity, so one threshold on t works for every later prefix. This proves the proposition. □

5 Missing composites

Proposition 7. *For every $\delta > 0$ there is t_0 such that, for all $t \geq t_0$ and every composite $h < E(t)$ outside U_t, $E(t) - h < \delta L(E(t))$.*

Proof. Fix $\delta > 0$, put $e = E(t)$, and suppose such a missing h has $D = e - h \geq \delta L(e)$. Take $t \geq 13$. Then $h \geq 15$, and h is odd because every positive even below e is used. Put $s = \mathrm{spf}(h)$. Equations (7) and (8) imply

$$D \leq 2sF(e) \leq 2BL(e), \qquad \alpha\sqrt{e} \leq s < \sqrt{e}, \quad \alpha = \frac{\delta}{2B}. \tag{16}$$

Set $\beta = \delta/(6B)$ and fix $M \geq 2$ with $M > 1/(2\alpha\beta)$. Select the even multiples $y = 2sMj$ in $(e - D/2, e)$. They all belong to U_t. Discard $a(t)$ if it belongs to this set, losing at most one value in total, and call the resulting set Y. Then

$$|Y| \geq \frac{D}{4sM} - 3. \tag{17}$$

Each $y = a(i_y)$ has a successor $u_y = a(i_y + 1)$ with $i_y + 1 \leq t$. The value h is still unused and eligible after y, so $u_y \leq h$ by the greedy rule. Equality is impossible since h is missing at time t. Thus $u_y < h$.

By (16), $y > e - BL(e) > e/2$ for large e. The old rate and a finite-prefix maximum force all i_y beyond any fixed cutoff, uniformly in y; in particular they exceed the cutoff for Proposition 6 with parameter $\delta/6$. For $f_y = E(i_y) \leq e$, that proposition gives

$$f_y \geq y - \frac{\delta}{6}L(f_y) \geq y - \frac{\delta}{6}L(e).$$

We used that L is increasing for arguments at least 3. Consequently,

$$f_y - u_y > \frac{D}{2} - \frac{\delta}{6}L(e) \geq \frac{\delta}{3}L(e). \tag{18}$$

These are post-occurrence frontiers. Their divergence follows from the uniform divergence of the indices i_y.

The successor $u = u_y$ is unused at time i_y and below f_y, so it is not even. It is not an odd prime p: its predecessor would be $2p$, whereas $y/2 = sMj$ is composite. It is not a tick, whose predecessor is odd, and it is not 1. It is therefore an ordinary odd composite. Both y and u are unchanged by smoothing. Their occurrence indices satisfy $e/4 < i_y < 2e$ for large e, by the old rate. Applying that rate at $i_y, i_y + 1$ gives

$$|u - y| \leq (6C + 1)L(e), \qquad e - A_B L(e) < u < e, \quad A_B = B + 6C + 1.$$

In particular $u \geq 15$ for sufficiently large e, before we apply the missing-composite bound to it. Combining (7), now at time i_y, with (18) yields

$$\frac{\delta}{3}L(e) < f_y - u \leq 2\,\mathrm{spf}(u)F(e), \qquad \mathrm{spf}(u) > \beta\sqrt{e}.$$

Every prime factor of u is large, while

$$j < \frac{\sqrt{e}}{2\alpha M} < \beta\sqrt{e},$$

and every prime factor of the fixed integer $2M$ is smaller than $\beta\sqrt{e}$ eventually. Since $u < e < (\beta\sqrt{e})^3$ eventually, Lemma 5 gives $u = sq_y$ with q_y prime. This does not assert that $s = \mathrm{spf}(u)$.

Distinct selected y have distinct successor indices and values, so the q_y are distinct. They lie in $((e - A_B L(e))/s, e/s)$, an interval of length $A_B L(e)/s$. By (17) and Lemma 4,

$$\frac{D}{4sM} - 3 \le \eta \frac{A_B L(e)}{s} + C_\eta.$$

Fix $\eta = \delta/(16MA_B)$. Since $s < \sqrt{e}$,

$$D \le \frac{\delta}{4} L(e) + 4M(C_\eta + 3)\sqrt{e} < \delta L(e)$$

for sufficiently large e, a contradiction. The choices have the order δ, the upper-excursion cutoff, M, η, C_η, and finally the frontier and index cutoffs. All are uniform in the missing composite h and the selected y. □

6 Exact counting and the frontier clock

We give the boundary terms explicitly, since the main coefficient arises from a difference of prime counts. Fix $t \ge 9$ with $e = E(t) \ge 14$, and put

$$\begin{aligned}
H &= \#\{h < e : h \text{ composite},\ h \notin U_t\},\\
O &= \#\{w \in U_t : w > e,\ w \text{ odd and not a tick}\},\\
P_1 &= \#\{p \text{ prime} : e/2 \le p < e\},\\
P_2 &= \#\{p \text{ prime} : e/3 < p < e/2\},\\
\theta_1 &= \mathbf{1}_{\{\exists \text{ odd prime } p<e/2 \text{ with } N_p=t\}},\\
\theta_2 &= \#\{p \text{ prime} : e/3 < p < e/2,\ N_p \in \{t-1, t\}\}.
\end{aligned}$$

Let $\theta = \theta_1 + \theta_2$; the spacing of introduction indices gives $0 \le \theta \le 2$.

For $p < e/2$, $2p < e$ is already used, so (9) shows that p is missing precisely when $N_p = t$. For $e/2 \le p < e$, $2p$ has not been used, so p is missing. For the first assertion the prime 2 is already used and is excluded. For the second, a used $2p \ge e$ would have entered at a frontier $2p = E(N_p - 1) \le e$, forcing the impossible used value e. There are therefore $H + P_1 + \theta_1$ missing positive integers below e. The used ticks above e are exactly those indexed by P_2, except for the θ_2 unfinished blocks. No used prime lies above e. Counting the t used values below and above the unused value e gives

$$t = (e - 1) - H - P_1 - \theta_1 + K + O + P_2 - \theta_2.$$

Write $I = \mathbf{1}_{\{e/2 \text{ prime}\}}$. Since e is even and at least 14,

$$P_1 = \pi(e) - \pi(e/2) + I, \qquad P_2 = \pi(e/2) - I - \pi(e/3).$$

We obtain the exact identity

$$\begin{aligned} e - t = \pi(e) - 2\pi(e/2) + \pi(e/3) \\ + H - K - O + 1 + \theta + 2I. \end{aligned} \tag{19}$$

Haver proves this counting identity in Corollary 12(b). Haver's Lean archive already proves `clock_balance`, `clock_identity`, `clock_identity_with_ticks`, and `clock_identity_from_interval_counts`; the last takes the endpoint errors as hypotheses. We discharge those hypotheses by proving that the missing-prime count is $P_1 + \theta_1$ and the tick count is $P_2 - \theta_2$, and by evaluating the interval endpoints through I and θ as above.

The old missing-composite estimate suffices to count holes. For $t \geq 13$, classify a missing composite by $s = \mathrm{spf}(h)$. By (7), it is an odd multiple of the odd prime s in $[e - 2sF(e), e)$. There are at most $F(e) + 1$ such multiples, because their spacing is $2s$. Hence

$$H \leq \pi(\sqrt{e})(13\pi(\sqrt{e}) + 725) = O(e/\log^2 e) = o(L(e)).$$

Equation (5) gives $K = o(L(e))$. Proposition 6 gives $O \leq \max\{T(t) - e, 0\}/2 + 1 = o(L(e))$. The correction $1 + \theta + 2I$ is between 1 and 5.

The prime number theorem, in the form $\pi(x) \sim x/\log x$ [4, 3], implies for each fixed $c > 0$ that $\pi(ce)/L(e) \to c$. Thus

$$\frac{\pi(e) - 2\pi(e/2) + \pi(e/3)}{L(e)} \longrightarrow 1 - 2 \cdot \frac{1}{2} + \frac{1}{3} = \frac{1}{3}.$$

Equation (19) yields $e - t = L(e)/3 + o(L(e))$. Since $L(e) = o(e)$ and $e \to \infty$, this implies $e \sim t$, and consequently $L(e) \sim L(t)$. We have proved

$$E(t) = t + \frac{1}{3}L(t) + o(L(t)). \tag{20}$$

The bound $O = o(L(e))$ proves the remaining hypothesis of the equivalence stated in Haver's Corollary 2(e) and proved in Corollary 12(d); the counting criterion itself is not new here.

7 Passage to every index

Proof of Theorem 1. First suppose $x = a(n)$ is ordinary, at a sufficiently large index, and use the pre-occurrence frontier $f = E(n-1)$. If $x < f$, then x is odd, unused at time $n - 1$, and neither 1 nor a prime, so Proposition 7 gives $f - x = o(L(f))$, uniformly over these occurrences. If $x = f$, the discrepancy is zero. If $x > f$, the unused even f remains unused at time n, whence $E(n) = f$. Now $x \leq T(n)$ and Proposition 6 gives $x - f = o(L(f))$. All three cases therefore give

$$a(n) = E(n-1) + o(L(E(n-1))).$$

By (20), $E(n-1) \sim n$, $L(E(n-1)) \sim L(n)$, and $E(n-1) = n + L(n)/3 + o(L(n))$. This proves the desired formula at all ordinary positions. The use of the pre-occurrence frontier also covers a step at which the frontier jumps.

For a prime block with introduction index $m = N_p$, all three smoothed values equal $2p = E(m-1)$. For $n = m, m+1, m+2$, the index $m-1$ is one of $n-1, n-2, n-3$. Equation (20) is unchanged on the $L(n)$ scale by any of these bounded shifts, since $L(n-j)/L(n) \to 1$ and $j = o(L(n))$. The formula follows at the prime and tick positions too.

For completeness, the uniform little-o conclusions here give exactly the quantifiers in (3). Given $\varepsilon > 0$, apply the two excursion propositions with $\delta = \varepsilon/8$. Eventually $L(E(n-1)) \leq 2L(n)$, so their contribution is at most $\varepsilon L(n)/4$. Take a common cutoff for the clock estimates shifted by $1, 2, 3$, each with error at most $\varepsilon L(n)/2$, and for the finite seed exclusions. Their sum is at most $\varepsilon L(n)$ at every ordinary index; the shifted clock alone covers the prime blocks. Increasing the common cutoff to at least 3 proves the theorem. □

8 Lean formalization and verification

The project `lean/` uses Lean 4 [2] and mathlib [8]. The complete chain from the literal greedy rule (1), through the smoothing (2), to Theorem 1 is machine-checked. Haver's estimates in Proposition 3 are *imported proved theorems from Haver's Lean sources*, recompiled in this development. They are neither independently re-proved mathematical contributions of this paper nor hypotheses or added axioms. The archive provides, among others, `EKG/Canonical.lean`, `EKG/EarlyEvens.lean`, `EKG/Span.lean`, `EKG/MissingComposites.lean` and `EKG/SmoothedPositions.lean`. Our new proofs of the two little-o estimates, the endpoint hypotheses of the counting identity, and the final transfer close the chain. The elementary constants used to weaken Haver's bounds are also proved.

The type transliterated beside Theorem 1 is independently fixed by `EndpointStatement` in `Statement.lean`. `Check.lean` requires `ekg_sharp_all_index` to have that type. For the rule, `actualA_properties`, `actualA_gcd_rule` and `actualA_gcd_minimal` prove positivity, non-repetition, gcd eligibility, and leastness, respectively. Thus the internal `GreedySeq` interface is instantiated by the recursively defined sequence, rather than left as an assumption. The proof sources accompany this manuscript in the same deposit.

The formal $a(n)$ is the zero-based canonical sequence evaluated at $n-1$; its value at $n = 0$ is a total extension unused in the theorem. The canonical construction proves existence, minimality, eligibility, and distinctness for the literal rule (1), including the prime factor 2. The new `smoothValue` definition excludes 2 in both branches. The bridge to Haver's smoothing is proved for every $n \geq 5$, using that the exceptional raw values $2, 6$ occur only at indices $2, 4$.

The files under `EKG/` correspond to the proof as follows:

`PartA.lean`, `PartB.lean`	upper and lower excursions
`CountingExact.lean`	exact counting identity
`EndpointRemainders.lean`	negligible remainders
`EndpointClock.lean`	prime counts and frontier clock
`EndpointStructure.lean`	ordinary terms and prime blocks
`EndpointTransfer.lean`	all-index transfer

The weak prime number theorem is obtained from `pi_alt'` in the pinned *PrimeNumberTheoremAnd* project [10]; its real-floor inclusive prime count is adapted in `SolutionPNT.lean`.

The Lean toolchain and dependency revisions are pinned in `lean/lean-toolchain` and `lean/lake-manifest.json`. Source provenance is recorded in `THIRD_PARTY.md`.

One inherited finite `native_decide` proof is replaced by ordinary kernel proofs; two compatibility edits change tactic lines only, without changing definitions or declaration types.

The exact output of `#print axioms ekg_sharp_all_index` is

```
'ekg_sharp_all_index' depends on axioms: [propext, Classical.choice, Quot.sound]
```

These are the usual Lean logical axioms, with no mathematical hypothesis added for an estimate, the counting identity or the prime number theorem. The imported PNT file contains unrelated admitted declarations `prelim_decay_2`, `prelim_decay_3` and `decay_alt`; none occurs in the proof of Theorem 1. This does not certify the whole upstream project. The three-branch corollary is the elementary algebraic consequence proved in the text; its separate statement is not included as an additional Lean theorem.

No human expert has reviewed it.

9 Numerical illustration

Table 1 gives the normalized error $R(n) = (b(n)-n)\log n/n$, whose limit is $1/3$ by Theorem 1. These values illustrate the formula; they are not used in its proof and do not imply a monotone approach to the limit.

n	$b(n)$	$R(n)$	$1/3$
1,000	1,043	0.297033477	0.333333333
10,000	10,361	0.332493287	0.333333333
100,000	103,088	0.355519138	0.333333333
1,000,000	1,024,686	0.341049694	0.333333333
10,000,000	10,210,804	0.339775904	0.333333333
100,000,000	101,839,957	0.338932605	0.333333333
500,000,000	508,457,544	0.338811220	0.333333333

Table 1: The smoothed EKG error, with absolute rounding error at most $5 \cdot 10^{-10}$ in each displayed ratio.

The included `data/ekg-table.cpp` generates exact integer terms using a bitmap and one least-unused-multiple cursor per prime, as in the method of LRS Section 3. Every eligible

value is a multiple of a prime divisor of the preceding term, so the minimum of these cursors gives the next value. The storage bound is checked at every selection; exhausting it aborts rather than truncating the candidate search. A separate direct gcd scan validates the first 4096 terms and rejects an altered prefix. All first 4096 smoothed values are also checked by trial division. The table file records 18 selected indices through $5 \cdot 10^8$, with integer digests in `data/ekg-table-summary.json`.

The ratios were certified by bounding $\log n$ with exact rational sums for $2 \operatorname{atanh} z$. After reducing the argument to $[1, 2]$, one uses 60 positive terms with $0 \le z \le 1/3$ and tail at most $2z^{121}/(121(1 - z^2))$. This certifies an absolute error at most 10^{-17} in the recorded ratios and correct rounding to nine decimal places in Table 1.

10 Related work

The argument completes the second-order programme of Haver [5], using the structural work of Lagarias, Rains, and Sloane [7] and Hofman and Pilipczuk [6]. Haver's Theorem 1(d) and Corollary 12(d) already prove the upper half of the frontier clock with sharp constant 1/3. The new contribution is the lower half plus the all-index transfer. Haver also proves the exact counting identity and the equivalence stated in Corollary 2(e), and identifies the rough-predecessor obstruction in the remark after Corollary 12. We discharge the remaining hypothesis of that equivalence and prove the uniform missing-composite estimate needed for the transfer, extending the Lean proof to the full second-order theorem. The prime-capacity lemma is an elementary fixed-modulus sieve bound [1], proved here; it does not require a short-interval prime number theorem. Haver's cited manuscript is unrefereed.

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Licence

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Disclosure

The main theorem is formally proved in Lean. No human expert has reviewed it.

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